Statistical Foundation for Data Analysis and Machine Learning

Comprehensive Training Notebook

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Abstract

This presentation provides a comprehensive guide to Correlation and Regression Analysis, transitioning from exploratory bivariate relationships to advanced predictive modeling and econometric theory.

- **Foundations:** Historical context, correlation vs. causation, and matrix algebra.
- **Modeling:** Simple and Multiple Linear Regression (SLR & MLR) using macroeconomic and clinical data.
- **Diagnostics:** Rigorous testing of regression assumptions (Normality, Homoscedasticity, Multicollinearity).
- **Econometric Applications:** Modeling macroeconomic inflation (CPI), estimating Marginal Elasticities via log-log models, and testing the Keynesian Consumption Function.

We emphasize the translation of statistical outputs into actionable, economically sound research reporting.

Session Objectives

- Distinguish between correlation and causation in observational data.
- Compute and interpret Pearson and Spearman correlation coefficients.
- Formulate, estimate, and interpret SLR and MLR models.
- Diagnose regression violations using formal statistical tests (Jarque-Bera, Breusch-Pagan, VIF).
- Apply log-log regression to macroeconomic data to interpret marginal elasticities and inflation drivers.
- Understand and estimate the Keynesian Marginal Propensity to Consume (MPC) using time-series data.

Part 1: Foundations of Regression

"The historical and theoretical underpinnings."

History of Regression Analysis

- **Francis Galton (Late 1800s):** Coined the term "regression" while studying the heights of parents and their children. He observed that extreme heights in parents tended to "regress toward the mean" in their offspring.
- **Karl Pearson & Ronald Fisher (Early 20th Century):** Formalized the mathematical foundations of correlation and regression, integrating them into the broader framework of inferential statistics and experimental design.
- **Modern Era:** Regression has evolved from a purely statistical tool to a foundational algorithm in machine learning, econometrics, and biostatistics.

Correlation vs. Causation

The Golden Rule of Statistics

Correlation does not imply causation.

- **Correlation:** A statistical measure describing the degree to which two variables move together. It quantifies *association*.
- **Causation:** Implies that a change in one variable *directly produces* a change in another.
- **The Trap:** Two variables may be highly correlated due to:
  1. Direct causation (A causes B).
  2. Reverse causation (B causes A).
  3. A hidden confounding variable (C causes both A and B).
  4. Pure coincidence (Spurious correlation).

Part 2: Correlation Analysis

"Quantifying the strength of association."

The `mtcars` Dataset

- Extracted from the 1974 *Motor Trend US* magazine.
- Contains fuel consumption and 10 aspects of automobile design and performance for 32 automobiles.

Variable	**Description**	**Type**
`mpg`	Miles/(US) gallon	Continuous
`wt`	Weight (1000 lbs)	Continuous
`hp`	Gross horsepower	Continuous
`cyl`	Number of cylinders	Categorical/Ordinal
`am`	Transmission (0=auto, 1=manual)	Binary

Pearson Correlation Analysis

- Measures the **linear** relationship between two continuous variables.
- Ranges from -1 (perfect negative) to +1 (perfect positive).

Variables	**Pearson's r**	**t-statistic**	**p-value**
`wt` & `mpg`	**-0.8676**	-9.559	1.29×10^{-10}

- **Interpretation:** A strong negative linear correlation. Heavier cars consistently achieve fewer miles per gallon.
- **Significance:** $p < 0.001$. We reject the null hypothesis of zero correlation.

Spearman Rank Correlation

- A non-parametric alternative to Pearson's r.
- Measures the **monotonic** relationship (based on ranked values) rather than strictly linear.
- **When to use:** When data is ordinal, contains severe outliers, or violates the normality assumption.

Variables	Spearman's ρ	Statistic (S)	p-value
`wt` & `mpg`	**-0.886**	10292	1.48×10^{-11}

- The Spearman correlation (-0.886) is slightly stronger than Pearson's, suggesting a highly consistent monotonic decrease in MPG as weight increases, robust to any extreme outliers.

Correlation Matrix

- Examines pairwise relationships between multiple continuous variables simultaneously.

	mpg	**cyl**	**disp**	**hp**	**wt**
mpg	1.00	-0.85	-0.85	-0.78	-0.87
cyl	-0.85	1.00	0.90	0.83	0.78
disp	-0.85	0.90	1.00	0.79	0.89
hp	-0.78	0.83	0.79	1.00	0.66
wt	-0.87	0.78	0.89	0.66	1.00

- **Insight:** `wt` and `disp` are highly correlated (0.89). Including both in a multiple regression model may cause multicollinearity.

Part 3: Simple Linear Regression (SLR)

"Modeling continuous outcomes with a single predictor."

SLR Concept & Equation

The Simple Linear Regression Model

$$Y = \beta_0 + \beta_1 X + \epsilon$$

- Y: Dependent variable (Outcome, e.g., `mpg`).
- X: Independent variable (Predictor, e.g., `wt`).
- β_0: Y-intercept (Value of Y when $X = 0$).
- β_1: Slope coefficient (Change in Y for a 1-unit increase in X).
- ϵ: Error term (Residuals, assumed to be normally distributed with mean 0).

SLR Model Output (`mtcars`)

Predictor	Estimate	Std. Error	t value	Pr >t
(Intercept)	37.285	1.878	19.86	$< 2e-16$ ***
`wt`	-5.344	0.559	-9.56	$1.29e-10$ ***

Interpretation

- **Slope ($\beta_1 = -5.344$):** For every 1,000 lbs increase in weight, MPG decreases by 5.344 units, holding all else constant.
- **Significance:** $p < 0.001$. We reject H_0. Weight is a highly significant predictor of fuel efficiency.

SLR R-Squared & Confidence Intervals

- **Multiple R-squared (0.7528):** 75.28% of the variance in MPG is explained by the vehicle's weight.
- **F-statistic:** 91.38 on 1 and 30 DF, $p < 0.001$.

Predictor	2.5% CI	97.5% CI
(Intercept)	33.45	41.12
`wt`	-6.49	-4.20

- We are 95% confident that the true effect of weight on MPG lies between -6.49 and -4.20. Because the interval **does not include zero**, it confirms statistical significance.

SLR Real-World Application: Hypertension Data

- **Dataset:** Clinical hypertension data (`htn_dat.csv`, $N = 4999$).
- **Objective:** Predict Systolic Blood Pressure (`SBP`) using Body Mass Index (`BMI`).

Predictor	Estimate	Std. Error	t value	p-value
(Intercept)	92.841	1.872	49.59	$< 2e-16$ ***
`BMI`	0.857	0.086	10.01	$< 2e-16$ ***

- **Equation:** $\text{SBP} = 92.84 + 0.857(\text{BMI})$
- **Interpretation:** For every 1 unit increase in BMI, SBP increases by 0.857 mmHg.
- **R-squared:** 0.041 (Only 4.1% of variance explained. Blood pressure is highly multifactorial!).

Part 4: Multiple Linear Regression (MLR)

"Controlling for confounders and improving model fit."

MLR Concept & Hypotheses

The Multiple Linear Regression Model

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_k X_k + \epsilon$$

- **Scenario:** Predicting `mpg` using both `wt` (weight) and `hp` (horsepower).
- **Hypotheses for Predictors:**
  - **H:** $\beta_i = 0$ (The predictor has no effect on Y, *holding other variables constant*).
  - **H:** $\beta_i \neq 0$ (The predictor has a significant effect).

MLR Model Output (`mtcars`)

Predictor	Estimate	Std. Error	t value	Pr >t
(Intercept)	37.227	1.599	23.28	$< 2e-16$ ***
`wt`	-3.878	0.633	-6.13	$1.12e-06$ ***
`hp`	-0.032	0.009	-3.52	0.00145 **

Interpretation

- **Weight (`wt`):** A 1,000 lb increase decreases MPG by 3.88 units, *holding horsepower constant*.
- **Horsepower (`hp`):** A 1 unit increase in HP decreases MPG by 0.032 units, *holding weight constant*.
- Both predictors are statistically significant ($p < 0.05$).

MLR Model Fit (R-Squared)

- **Multiple R-squared:** 0.8268
- **Adjusted R-squared:** 0.8148

Interpretation

By adding `hp` to the model, the explained variance in MPG increased from 75.28% (SLR) to 82.68% (MLR).

The **Adjusted R-squared** (0.8148) is the preferred metric for MLR, as it penalizes the addition of irrelevant predictors. It indicates that 81.48% of the variability in fuel efficiency is explained by weight and horsepower combined.

Part 5: Regression Diagnostics

"Ensuring the validity of our statistical inferences."

The Core Assumptions of Linear Regression

- Ordinary Least Squares (OLS) regression relies on strict assumptions. If violated, coefficients may be biased, or p-values invalid.

The "LINE" Assumptions

1. **L**inearity: The relationship between X and Y is linear.
2. **I**ndependence: Observations (residuals) are independent (no autocorrelation).
3. **N**ormality: The residuals are normally distributed.
4. **E**qual Variance (Homoscedasticity): The variance of residual errors is constant across all levels of X.

Diagnostic 1: Multicollinearity

- Occurs when independent variables are highly correlated with each other.
- **Consequence:** Inflates standard errors, making coefficients unstable and statistically insignificant.
- **Detection:** Variance Inflation Factor (VIF).

Predictor	**VIF**
`wt`	1.77
`hp`	1.77

- **Rule of Thumb:** VIF > 5 (or 10) indicates severe multicollinearity. Here, VIF is low; no action needed.

Diagnostic 2: Influential Observations

- Some data points may exert disproportionate leverage on the regression line.
- **Detection:** Cook's Distance.
- **Threshold:** Observations with Cook's Distance $> \frac{4}{N}$ are considered highly influential.

Action Taken

In the `mtcars` MLR model, 4 influential cars were identified: *Chrysler Imperial, Fiat 128, Toyota Corolla, Maserati Bora*.

Best Practice: Do not blindly delete outliers. Investigate them first. If they are data entry errors, remove them. If they are genuine extreme cases, consider robust regression techniques.

Diagnostic 3: Normality & Homoscedasticity

- **Normality (Jarque-Bera Test):**
  - Tests whether residuals have skewness and kurtosis matching a normal distribution.
- **Homoscedasticity (Breusch-Pagan Test):**
  - H: Residuals have constant variance.
  - If p-value ¡ 0.05, we reject H_0 and conclude heteroscedasticity is present.
- **Independence (Durbin-Watson & Breusch-Godfrey Tests):**
  - Tests for autocorrelation in the residuals, common in time-series data.

Part 6: Macroeconomic Modeling

"Predicting inflation using macroeconomic indicators."

Macroeconomic Model Context

- **Dataset:** Annual macroeconomic indicators.
- **Target Variable:** Consumer Price Index (`CPI`) - A measure of average price changes and inflation.
- **Predictors:**
  - **Exchange Rate (`ExchRate`):** Local currency per USD.
  - **Lending Interest Rates (`LendIntRates`):** Cost of borrowing.

Economic Theory

In an open economy, inflation is driven by **imported inflation** (exchange rate depreciation) and **demand-pull factors** (influenced by interest rates and money supply).

Macroeconomic Model Output

Predictor	Estimate	Std. Error	t value	p-value
(Intercept)	7.900	15.023	0.526	0.604
`ExchRate`	1.266	0.158	8.011	< 0.001 ***
`LendIntRates`	-1.486	0.510	-2.912	0.008 **

- **R-squared:** 0.772 (77.2% of inflation variance explained).
- **F-statistic:** 38.96, $p < 0.001$. The overall model is highly significant.

Macroeconomic Interpretation

Exchange Rate (Coefficient = 1.266)

A 1 unit depreciation in the exchange rate increases the CPI by 1.266 units. This confirms the presence of **imported inflation**: as the local currency weakens, the cost of imported goods rises, directly pushing up the consumer price index.

Lending Interest Rates (Coefficient = -1.486)

A 1% increase in lending rates decreases the CPI by 1.486 units. This reflects the **contractionary effect** of monetary policy: higher interest rates reduce borrowing, cool down aggregate demand, and ultimately ease price pressures.

Macroeconomic Policy Implications

- **Currency Stabilization:** Since the exchange rate is the dominant driver of inflation (highest t-statistic), the Central Bank must prioritize foreign exchange reserves and currency stabilization to protect consumer purchasing power.
- **Monetary Policy Transmission:** The negative coefficient for interest rates proves that the Central Bank's rate hikes effectively cool inflation. However, policymakers must balance this against the risk of stifling economic growth.
- **Fiscal Coordination:** With 77.2% of inflation explained by these two variables, the remaining 22.8% is likely driven by supply-side shocks (e.g., fuel prices, agricultural output) requiring targeted fiscal interventions.

Transition

Part 7: Marginal Elasticities

"Modeling percentage changes and economic pass-through."

The Log-Log (Double-Log) Regression Model

The Econometric Specification

$$\ln(Y) = \beta_0 + \beta_1 \ln(X_1) + \beta_2 \ln(X_2) + \epsilon$$

- **Why Log-Log?** In economics, we often care about *percentage changes* rather than absolute unit changes.
- **Elasticity:** The coefficients (β_i) are interpreted as **marginal elasticities**.
- **Meaning:** A 1% increase in X leads to a β% change in Y, holding other factors constant.
- **Advantage:** Elasticities are unit-free, making them ideal for comparing effects across different macroeconomic indicators.

Elasticity Model Output (Inflation Drivers)

Predictor	Estimate	Std. Error	t value	p-value
(Intercept)	-0.704	0.528	-1.335	0.195
`log(ExchRate)`	1.515	0.085	17.758	< 0.001 ***
`log(LendIntRates)`	-0.552	0.138	-4.014	< 0.001 ***

- **Multiple R-squared:** 0.9348 (93.48% of CPI variation explained).
- **Adjusted R-squared:** 0.9291 (Very close to R^2, indicating no overfitting).
- **F-statistic:** 164.8, $p < 0.001$. The model is highly significant.

Interpreting Marginal Elasticities

Exchange Rate Elasticity ($\beta_1 = 1.515$)

A 1% depreciation in the exchange rate is associated with a **1.515% increase** in CPI. Since $1.515 > 1$, inflation is **elastic** with respect to the exchange rate. This indicates a severe *exchange rate pass-through* to domestic prices.

Lending Interest Rate Elasticity ($\beta_2 = -0.552$)

A 1% increase in lending interest rates is associated with a **0.552% decrease** in CPI. Since $|-0.552| < 1$, the effect is **inelastic**. Higher rates reduce borrowing and aggregate demand, easing inflation, but the percentage impact is less than proportional.

Economically Sound Reporting (Elasticities)

- **Research Paper Synthesis:** "The double-log model reveals highly elastic exchange rate pass-through, meaning currency depreciation disproportionately fuels inflation. Conversely, monetary tightening (lending rates) has an inelastic but significant dampening effect."
- **What we cannot compute here:** Unlike the linear consumption model, we **cannot** compute the Keynesian multiplier ($k = \frac{1}{1-MPC}$) from this model.
- **Why?** The coefficients are elasticities (percentage responses), not Marginal Propensities to Consume (absolute dollar responses).
- **Takeaway:** Always match your mathematical interpretation to the functional form of your econometric model.

Part 8: Keynesian Economics & Consumption

"Estimating the Marginal Propensity to Consume."

The Keynesian Consumption Function

The Theory

Coined by **John Maynard Keynes** in his seminal 1936 work, *"The General Theory of Employment, Interest and Money"*.

- **Absolute Income Hypothesis:** Consumption depends primarily on current disposable income.
- **Fundamental Psychological Law:** "Men are disposed, as a rule and on average, to increase their consumption as their income increases, but not by as much as the increase in their income."
- **Mathematical Representation:** $C = \alpha + \beta Y$
  - C: Consumption
  - Y: Income
  - α: Autonomous consumption (consumption when income is zero)
  - β: **Marginal Propensity to Consume (MPC)**

Marginal Propensity to Consume (MPC)

Definition

The MPC is the proportion of an additional unit of income that is spent on consumption.

$$MPC = \frac{\Delta C}{\Delta Y}$$

- **Theoretical Bounds:** $0 < MPC < 1$. (People spend a fraction of new income and save the rest).
- **The Multiplier Effect:** The Keynesian multiplier is defined as $k = \frac{1}{1-MPC}$. A higher MPC means fiscal stimulus (e.g., government spending or tax cuts) will have a larger magnified effect on total economic output (GDP).

Consumption Model Context

- **Dataset:** Historical macroeconomic time-series data (1950–1985).
- **Variables:**
  - **Dependent (`Consumption`):** Aggregate personal consumption expenditures (in billions).
  - **Independent (`Income`):** Aggregate personal income (in billions).

Objective

Estimate the empirical consumption function and extract the MPC to understand the structural dynamics of this economy.

Consumption Model Output

Predictor	Estimate	Std. Error	t value	p-value
(Intercept)	11.374	9.629	1.181	0.246
`Income`	0.898	0.006	153.603	$< 2e-16$ ***

- **R-squared:** 0.9986 (99.86% of the variance in consumption is explained by income).
- **F-statistic:** 23,594, $p < 0.001$.
- **Equation:** Consumption $= 11.37 + 0.898 \times$ Income

Interpreting the MPC

Empirical Estimate

The slope coefficient for Income is **0.898**. Therefore, the empirical **MPC = 0.898**.

- **Economic Meaning:** For every additional $1 billion of income generated in the economy, consumers spend $898 million and save $102 million.
- **Validation of Keynes:** The MPC is strictly between 0 and 1, perfectly validating Keynes's fundamental psychological law.
- **Autonomous Consumption:** The intercept (11.37) represents baseline consumption required for survival, even if aggregate income drops to zero (funded by debt or drawing down savings).

Policy Implications of the MPC

- **High MPC (0.898):** Indicates a highly consumer-driven economy where households live close to their means.
- **The Multiplier:** $k = \frac{1}{1-0.898} = \frac{1}{0.102} \approx 9.8$.
- **Fiscal Policy Impact:** A \$1 billion injection of government spending will theoretically generate \$9.8 billion in total GDP growth.
- **Tax Cuts vs. Spending:** Because the MPC is so high, tax cuts aimed at lower-income households (who have a higher MPC than the wealthy) will be extremely effective at stimulating aggregate demand.

Part 9: Synthesis & Best Practices

"Bridging the gap between statistical output and real-world decisions."

Best Practices for Economically Sound Reporting

1. **Visualize First:** Always plot scatter plots and correlation heatmaps before fitting models.
2. **Check Assumptions:** Test for LINE assumptions. Address autocorrelation in time-series data.
3. **Match Interpretation to Functional Form:** Report *multipliers* for linear models and *elasticities* for log-log models. Never mix them up.
4. **Economic Significance vs. Statistical Significance:** A tiny p-value does not mean the effect is economically meaningful. Check theoretical bounds (e.g., $0 < MPC < 1$).
5. **Comprehensive Reporting:** Always report R^2, Adjusted R^2, F-statistics, and Confidence Intervals alongside coefficient estimates to provide a complete picture of model fit and uncertainty.

Q&A and Thank You!

Questions?

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